

Customer Journey Map

2. REQUIREMENT ANALYSIS

Date	4 July 2026
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well Being
Maximum Marks	3 Marks

Technology Stack Specification

This document lists and details the core technologies chosen for the HDI Predictor implementation, justifying their selection based on project constraints.

✂ Technology Stack List

TECHNOLOGY LAYER	CHOSEN COMPONENT	VERSION / CONTEXT	SELECTION JUSTIFICATION
Programming Language	Python	3.10+	Core language for data science and web scripting with massive package ecosystem.
Backend Framework	Flask	3.0+	Lightweight WSGI web framework suited for rapid development and API routing without ORM overhead.

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Machine Learning	Scikit-Learn	1.2+	Standard ML library containing efficient implementation of Linear Regression and scaling tools.
Data Manipulation	Pandas	2.0+	Provides powerful DataFrame manipulation routines used for parsing wide-format UNDP datasets.
Scientific Operations	NumPy	1.24+	Drives linear algebra calculations and array standardizations behind ML predictions.
Visualizations	Matplotlib / Seaborn	3.7+ / 0.12+	Generation of high-fidelity EDA plots (correlation matrices, KDE distributions, scatters).
Frontend Framework	Bootstrap 5	5.3.0	Provides responsive layout grid system, pre-built utility forms, and buttons.
Icons Library	Font Awesome	6.4.0	Comprehensive icons list mapping specific development areas (health, cap, line chart).
Client Scripting	Vanilla JS	ES6 / Native	Lightweight scripting used to control range syncs, theme toggles, and localStorage caching without heavy framework binaries.
Persistence	Browser LocalStorage	HTML5 standard	Eliminates database integration, offering client-side history logs without backend data leaks.

System Runtime Context

- **Development WSGI:** Flask Built-in server (debug mode active for trace logs).
- **Production WSGI Recommendation:** Gunicorn or Waitress.

- **MathJax CDN:** Standard javascript CDN used to render LaTeX calculations (MathJax-script) directly in /about routes.